HONGBO WEN

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I am a Ph.D. candidate in Computer Science at UC Santa Barbara, advised by Yu Feng. My dissertation, *Soundness and Correctness of Computation over Diverse Algebraic Systems*, develops program analysis, synthesis, and testing techniques for preserving specification-level guarantees across finite fields, fixed-point arithmetic, cryptographic groups, and floating-point kernels. My broader research applies specification-driven security to LLM agents and AI-assisted vulnerability discovery. This work has uncovered zero-day vulnerabilities across proof systems, financial protocols, and agent platforms, has been supported by the **Ethereum Foundation**, and has involved collaborations with **Google Security** and **a16z crypto**.

EDUCATION

University of California, Santa Barbara, CA, USA Sep 2022 - Present
Ph.D. candidate in Computer Science (expected 2027)
GPA 4.0/4.0

Tsinghua University, Beijing, China Aug 2015 - Jun 2020
Bachelor of Engineering

SELECTED PUBLICATIONS

[1] Ying Li, Yanju Chen, **Hongbo Wen**, Bosi Zhang, Hanzhi Liu, Peiran Wang, Yu Feng, Yuan Tian. *VIGIL: Runtime Enforcement of Behavioral Specifications in AI Agent Skills.* **arXiv 2026**.

[2] Ying Li, **Hongbo Wen**, Yanju Chen, Hanzhi Liu, Yuan Tian, Yu Feng. *No Attack Required: Semantic Fuzzing for Specification Violations in Agent Skills.* **arXiv 2026**.

[3] **Hongbo Wen**, Ying Li, Hanzhi Liu, Chaofan Shou, Yanju Chen, Yuan Tian, Yu Feng. *SEMIA: Auditing Agent Skills via Constraint-Guided Representation Synthesis.* **arXiv 2026**.

[4] Hanzhi Liu, Chaofan Shou, Xiaonan Liu, **Hongbo Wen**, Yanju Chen, Ryan Jingyang Fang, Yu Feng. *Synthesizing Multi-Agent Harnesses for Vulnerability Discovery.* **arXiv 2026**.

[5] Hanzhi Liu, Chaofan Shou, **Hongbo Wen**, Yanju Chen, Ryan Jingyang Fang, Yu Feng. *Your Agent Is Mine: Measuring Malicious Intermediary Attacks on the LLM Supply Chain.* **arXiv 2026**.

[6] Junrui Liu, Jiaxin Song, Yanning Chen, Hanzhi Liu, **Hongbo Wen**, Luke Pearson, Yanju Chen, Yu Feng. *Tabby: A Synthesis-Aided Compiler for High-Performance Zero-Knowledge Proof Circuits.* **OOPSLA 2025**.

[7] **Hongbo Wen**, Hanzhi Liu, Yanju Chen, Jingyu Ke, Dahlia Malkhi, Yu Feng. *Thunderbolt: Fast Asynchronous Off-Chain Bitcoin Transfers.* **IACR ePrint 2025**.

[8] **Hongbo Wen**, Hanzhi Liu, Jiaxin Song, Yanju Chen, Wenbo Guo, Yu Feng. *FORAY: Towards Effective Attack Synthesis against Deep Logical Vulnerabilities in DeFi Protocols.* **ACM CCS 2024**.

[9] **Hongbo Wen**, Jon Stephens, Yanju Chen, Kostas Ferles, Shankara Pailoor, Kyle Charbonnet, Isil Dillig, Yu Feng. *Practical Security Analysis of Zero-Knowledge Proof Circuits.* **USENIX Security 2024**.

INVITED TALKS

Tabby: A Synthesis-Aided Compiler for High-Performance Zero-Knowledge Proof Circuits Feb 2026
SoCalPLS USC, Los Angeles, CA, USA

Tabby: Automated Programming of Efficient Zero-Knowledge Proof Circuits Feb 2025
ETHDenver Denver, CO, USA

Practical Security Analysis of Zero-Knowledge Proof Circuits Feb 2025
SoCalPLS UCSD, San Diego, CA, USA

FORAY: Towards Effective Attack Synthesis against Deep Logical Vulnerabilities in DeFi Protocols Oct 2024
ACM CCS '24 Salt Lake City, UT, USA

Practical Security Analysis of Zero-Knowledge Proof Circuits Aug 2024
USENIX Security '24 Philadelphia, PA, USA

SELECTED GRANTS

Ethereum Foundation Academic Grants Round 2024 Jun 2024
Yu Feng, Hanzhi Liu, Hongbo Wen
Topic: Sentinel: Adaptive Counter-Attack Synthesis for Mitigating Onchain Exploits

Ethereum Foundation Academic Grants Round 2023 Jun 2023
Yu Feng, Yanju Chen, Hongbo Wen
Topic: Financial Model-Driven Attack Synthesis for DeFi

SELECTED ZERO-DAY DISCOVERIES

[1] **LLM Agents**: Filed **50+ zero-day** security advisories on OpenClaw/ClawHub agent skills, covering credential leakage, RCE, PII exfiltration, and indirect prompt injection.
[2] **DeFi**: Discovered **10 zero-day** vulnerabilities on BNB Chain DeFi protocols using *FORAY*, with verified exploit traces matching $\geq \$21M$ historical-loss patterns.
[3] **ZK Circuits**: Discovered zero-day vulnerabilities in widely-used Circom projects using *ZKAP*, including **0xPARC, Polygon-ZK, Axiom, iden3, privacy-ethereum**.

WORK EXPERIENCE

Riema Labs Inc. Jan 2024 - May 2026
Technical Advisor.
Highlights:

- Advised a **50+** engineering team on shipping security-critical production systems and AI-agent native engineering transformation, serving **150K+** users.
- Shipped cross-chain bridges, Bitcoin-native ZK infrastructure, and user-custody wallet systems.

Veridise Inc. Mar 2022 - Aug 2022
Research & Development Engineer: smart contract and zero-knowledge proof security.
Highlights:

- Discovered **30+** critical vulnerabilities in widely-used ZK projects, and **10+** zero-day DeFi vulnerabilities.

SERVICE

USENIX Security '25 Artifact Evaluation Committee 2025
CS162 Programming Languages, Teaching Assistant, UC Santa Barbara 2023 – 2025

SKILLSET

Languages: Rust, Go, Solidity, C/C++, JavaScript, Python.
AI-Assisted Research: Several projects and publications herein were co-developed with LLM coding agents, directed and verified by me.